



Ontology For Attacks in Cyber Risk Assessment (OACRA)

Brian A. Haugh
Steven P. Wartik

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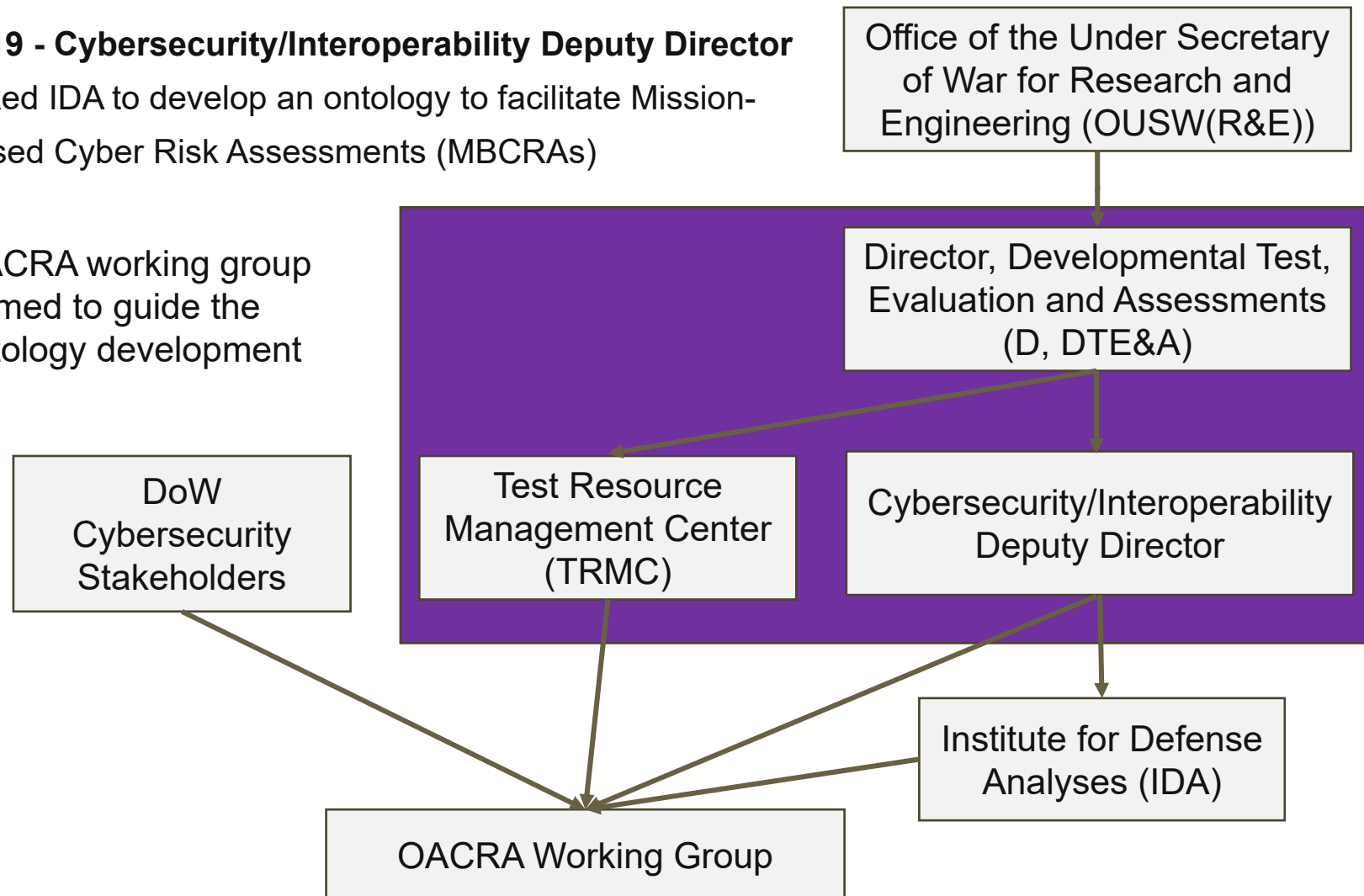
Institute for Defense Analyses

730 East Glebe Road • Alexandria, Virginia 22305

OACRA Origins

2019 - Cybersecurity/Interoperability Deputy Director asked IDA to develop an ontology to facilitate Mission-Based Cyber Risk Assessments (MBCRAs)

OACRA working group formed to guide the ontology development



OACRA Working Group Purposes

- **Promote** collaboration and information sharing between services and cyber DT&E organizations.
- **Develop** an ontology approach that captures technology-specific but program-agnostic Cyber Table Top (CTT)/MBCRA attacks that include engagement and presence methods, non-attributional effects, and command and control techniques.
- **Enhance** and increase the efficiency of CTT/MBCRA executions by providing the teams representing the opposing cyber force with readily available data on (potential) attacks previously identified.

OACRA Task Initiation

- Options for developing OACRA
 - Alternative possible sources of OACRA concepts
 - Integration options for different sources
 - Ontologies to use as baselines to start with at different levels
 - Upper-level
 - Mid-level
 - Domain
 - Different models of cyber-attacks
 - Alternative ontology development tools
 - Alternative knowledge base tools to support MBCRA ontologies and data

Initial Sources of MBCRA Concepts – Top-down

1. MITRE ATT&CK® Matrix¹
2. National Security Agency (NSA) Threat Framework
3. Lockheed Martin Seven Step Cyber Kill Chain²
4. Centralized Cyber Capabilities Directory (C3D) Lexicon³
5. Existing relevant ontologies
 - a. Common Core Cyber Ontology (C3O), from CUBRC, the Army Intelligence and Information Warfare Directorate (I2WD) and C3O WG
 - b. Damage Assessment Ontology (DAO), from IDA
 - c. Unified Cyber Ontology (UCO), a Linux Foundation project
 - d. National Information Exchange Model (NIEM), originally a government standard, now open source

1) MITRE ATT&CK and ATT&CK are registered trademarks of The MITRE Corporation. <https://attack.mitre.org/>.
https://media.defense.gov/2019/Jul/16/2002158108/-1/-1/0/CTR_NSA-CSS-TECHNICAL-CYBER-THREAT-FRAMEWORK_V2.PDF.

2) <https://www.lockheedmartin.com/en-us/capabilities/cyber/cyber-kill-chain.html>.

3) The C3D is a searchable directory of cyber T&E tools, facilities, service providers, and people that is hosted by the Joint Federated Assurance Center (JFAC). <https://jfacym.id/c3d>

OACRA Data Sources – Bottom-up Ontology Development

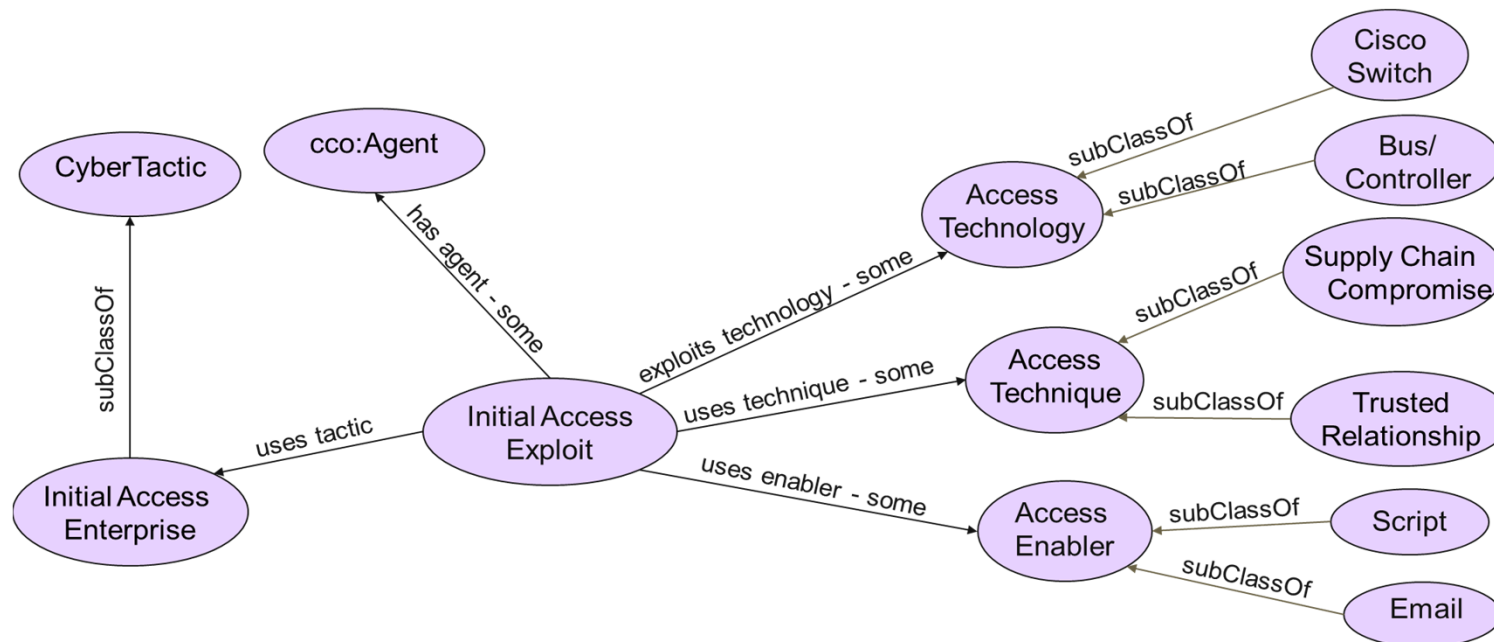
- Cyber Table Top (CTT) Exercises
 - Review possible cyber attacks against DoW systems, their possible outcomes, results, and mitigations
 - Use spreadsheet templates to gather exercise data

OPFOR					Control Team/OPFOR		Analysis Team
Goal	Attack Method	Attack Description	Assumptions	When in the Mission Timeline	Possible Outcome	Attack Result	Common Mitigations
ACCESS	SW Supply Chain Attack	Gain Access via modified application of firmware upload version 7.2.	Adversary able to obtain a baseline binary and has means to reverse engineer	During planned maintenance activity; occurs once a month from prime contractor facility	Server BIOS is uploaded with infected binary	Adversary has a presence in the HW BIOS on system A's servers.	Develop and Implement Patch Management Plan for SW XX
ACCESS	SW Supply Chain Attack	Gain Access via modified application of firmware upload version 7.2.	Adversary able to obtain a baseline binary and has means to reverse engineer	During planned maintenance activity; occurs once a month from prime contractor facility	Map Web Service has malicious browser exploit	Adversary has a presence on the application web server that disseminate payloads to onboard client machines	Perform supply chain assessment on primary SW vendor
C2	IFF Spoofing	Use IFF Spoofing to communicate with Implants onboard platform A	Adversary is able to obtain programmable IFF transponders	During platform A training operations	Adversary places transponder in environment that responds with specific modes 4 and 5	Adversary has low bandwidth C2 channel implant	Implement Dual Factor Authentication on Platform A Transponder A & B

Circled notional system-specific content removed to conceal sources.

OACRA Data Sources – “Technologies”

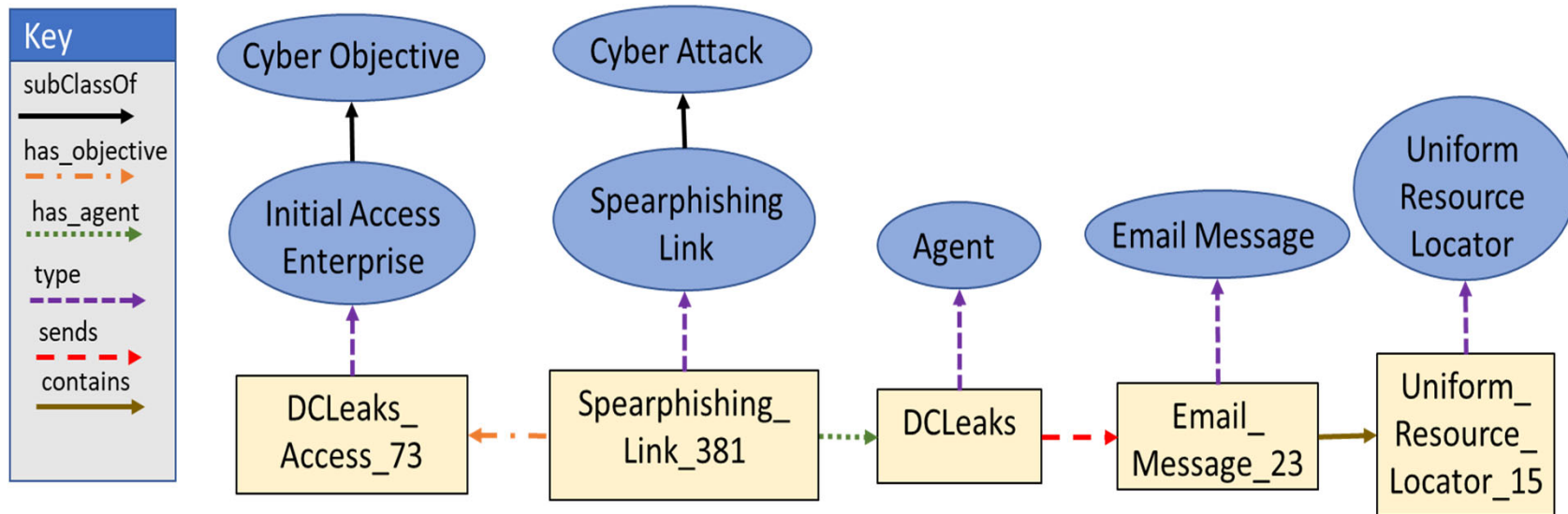
- Initial “Technologies” Concept Used in Cyber Attacks
 - Specific classes of “technologies” cited by participants in CTTs
 - Exploited technologies – e.g., switches, routers, buses, software, ..
 - Enabling technologies – e.g., scripts, email, malware, ...



Goal	Technique	Exploited Technology	Enabling Technology
Initial Access Enterprise	Supply Chain Compromise	Cisco Switch	Script

OACRA Ontology Concept Refinement

- Chose C3O/CCO/BFO as Cyber Domain Ontology
- Focused on MITRE ATT&CK for Cyber Attack Concepts
- Initial Extensions of C3O with Key MBCRA Concepts
 - *Cyber Attack*, *CyberObjective*, *Act of Cyber Threat Mitigation*, .
 - *Cyber Objective* – subClassOf cco:Objective, captures ATT&CK tactics¹
- Refined model of Cyber Attacks



¹MITRE ATT&CK defines a "tactic" as a "tactical goal."

"Frequently Asked Questions," accessed August 26, 2025, <https://attack.mitre.org/resources/faq/>.

Further OACRA Refinements

- Technologies re-categorized to Use Common Cyber Terms
 - Exploited technologies => Cyber Target
 - Enabling technologies => Attack Vector
- CTT Template Revisions
 - Added pull-down menus to select pre-defined options
 - Goal
 - MITRE ATT&CK Domain – Enterprise, Mobile, ICS
 - MITRE ATT&CK Tactic
 - MITRE ATT&CK Technique
 - Attack Vector – hierarchy of classes, e.g., ‘Information Artifact’
 - Attack Timing – generic categories of time (e.g., ‘Maintenance Period’)
 - Other template fields retained as text values
 - Attack Description, Assumption, Possible Outcome

Mapping CTT Template Fields to OACRA Classes

Goal	MITRE ATT&CK Domain	MITRE ATT&CK Tactic	MITRE ATT&CK Technique	Attack Description	Attack Vector Superclass	Attack Vector Class	Assumptions	Attack Timing	Possible Outcome or Possible System Impact
Access	Enterprise_Domain	Initial Access_Enterprise	Compromise Software Supply Chain (T1195.002)	Gain Access via modified application of firmware upload	Information_Artifact	Firmware	Adversary able to obtain baseline binary and has means to reverse engineer.	Maintenance Period	Server BIOS is uploaded with infected binary
Goal	MITRE ATT&CK Domain	MITRE ATT&CK Tactic	MITRE ATT&CK Technique	Attack Description	Attack Vector Superclass	Attack Vector Class	Assumptions	Attack Timing	Possible Outcome or Possible System Impact
'oacra: Access'	[Just narrows tactics and techniques in template]	'cco:Initial Access_Enterprise Tactic'	'cco: Compromise Software Supply Chain Enterprise Technique'	Gain Access via modified application of firmware upload @en	'oacra: Information Artifact'	cco:Firmware	Adversary able to obtain baseline binary and has means to reverse engineer. @en	'oacra: Maintenance Period'	Server BIOS is uploaded with infected binary @en

Key Maps to instance of class 

Maps to same text 

CTT Goal Field

- Sources
 - Guidebook for *DoD Cyber Table Top Exercise, Version 1*
 - *Centralized Cyber Capabilities Directory (C3D)*

The screenshot displays the Protégé ontology editor interface. On the left, the 'Class hierarchy: Access' is shown, with 'Access' highlighted under 'Cyber Method'. The hierarchy includes: Cyber Act, Cyber Kill Chain, Cyber Method (with sub-classes: Access, Analyze, Command and Control, Exfiltrate, Exploit, Forensics, Information Gathering, Obfuscate, Pivot, Sanitize, Visualize, Vulnerability Analysis), Cyber Attack, Cyber Embedded Technique, Cyber Enterprise Technique, Cyber ICS Technique, and Cyber Mobile Technique. The right pane shows the 'Access' class details, including its URI (<http://www.semanticweb.org/OACRA/Access>), annotations, and a description: 'Ability and means to communicate with or interact with a system, use system resources to handle information, gain knowledge of the information the system contains, or control system components/functions.' Below the description, it shows 'SubClass Of' relationships: 'Cyber Method' and 'has objective' some 'Cyber Objective'.

Cyber Attack Classes

- Sources – MITRE ATT&CK®, MITRE Embedded Systems
- Three matrices from ATT&CK®

The screenshot displays the Protege ontology editor interface. The top navigation bar includes tabs for 'Active ontology', 'Entities', 'Individuals by class', and 'DL Query'. The left sidebar shows a 'Class hierarchy' for 'Cyber Attack', listing various subclasses such as 'Act of Artifact Employment', 'Act of Decoy Use', 'Act of Facility Use', 'Act of Financial Instrument Use', 'Act of Legal Instrument Use', 'Act of Portion of Material Consumption', 'Act of Tool Use', 'Act of Vehicle Use', 'Act of Weapon Use', 'Cyber Act', 'Cyber Kill Chain', 'Cyber Method', 'Cyber Attack' (selected), 'Cyber Embedded Technique', 'Cyber Enterprise Technique', 'Cyber ICS Technique', and 'Cyber Mobile Technique'. The main panel shows the 'Annotations' tab for the 'Cyber Attack' class, with the following information:

- Annotations:** Cyber Attack
- Annotations:** +
- rdfs:label** [language: en] [in CyberOntology]: Cyber Attack
- rdfs:comment** [language: en] [in CyberOntology]: A malicious act is an intentional, wrongful act performed against another without legal justification or excuse, <https://definitions.uslegal.com/m/malicious>.
- definition** [language: en] [in CyberOntology]: An Act of Artifact Employment in which an Agent uses a computer to engage in a malicious act.
- is curated in ontology** [type: xsd:anyURI] [in CyberOntology]: <http://www.ontologyrepository.com/CommonCoreOntologies/Domain/CyberOntology>
- Description:** Cyber Attack
- SubClass Of** +
 - 'Act of Artifact Employment'
 - 'has agent' some (uses some Computer)

At the bottom, a status bar indicates: 'To use the reasoner click Reasoner > Start reasoner' and 'Show Inferences'.

Cyber Objective Class

- A general class including malign and benign objectives to be achieved by a Cyber Act

The screenshot displays the OACRA ontology editor interface. The top menu bar includes File, Edit, View, Reasoner, Tools, Refactor, Window, and Help. The address bar shows the URL: <http://www.semanticweb.org/2023-04-11/OACRA>. The breadcrumb trail indicates the path: > entity > continuant > generically dependent continuant > Information Content Entity > Directive Information Content Entity > Objective > Cyber Objective.

The interface is divided into several panes. On the left, the 'Class hierarchy: Cyber Objective' pane shows a tree structure of classes. The 'Cyber Objective' class is selected, and its subclasses are listed: Cyber Embedded Tactic, Cyber Enterprise Tactic, Collection Enterprise Tactic, Command and Control Enterprise Tactic, Credential Access Enterprise Tactic, Defense Evasion Enterprise Tactic, Discovery Enterprise Tactic, Execution Enterprise Tactic, Exfiltration Enterprise Tactic, Impact Enterprise Tactic, Initial Access Enterprise Tactic, Lateral Movement Enterprise Tactic, Persistence Enterprise Tactic, Privilege Escalation Enterprise Tactic, Reconnaissance Enterprise Tactic, Resource Development Enterprise Tactic, Cyber ICS Tactic, and Cyber Mobile Tactic.

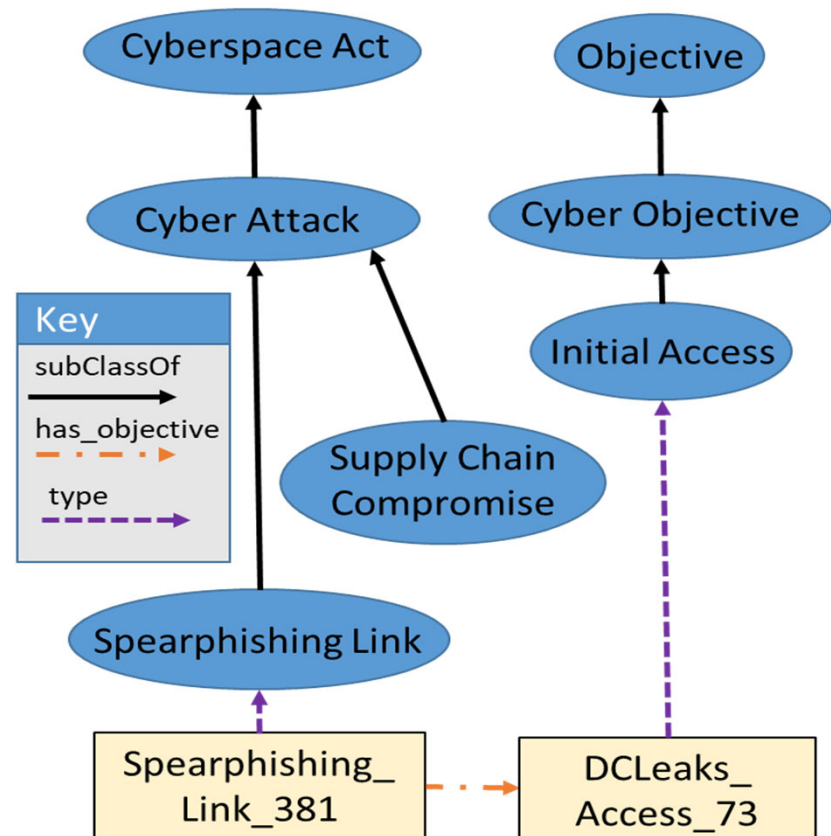
The main pane displays the 'Annotations: Cyber Objective' section. It lists several annotations for the 'Cyber Objective' class:

- rdfs:label** [language: en] [in CyberOntology]: Cyber Objective
- rdfs:comment** [language: en] [in CyberOntology]: This class was added to the Version 1.3 of the Cyber Ontology created for the Ontology for Attacks in Cyber Risk Assessments (OACRA), although its inclusion has not yet been vetted by the Cyber Ontology Working Group.
- dc:source** [language: en] [in CyberOntology]: Derived from Webster's Dictionary definition for "objective"
- definition** [language: en] [in CyberOntology]: A purpose to be achieved by a cyber act or operation.
- is curated in ontology** [language: en] [in CyberOntology]: <http://www.ontologyrepository.com/CommonCoreOntologies/Domain/CyberOntology>

The 'Description: Cyber Objective' pane at the bottom shows the 'SubClass Of' relationship, indicating that 'Cyber Objective' is a subclass of 'Objective'.

Relation between Cyber Attack and Cyber Objective

- Initial concept `uses_tactic` relating a Cyber Attack to a Tactic was replaced by `has_objective`
- Property that acknowledges the definition of an ATT&CK® Tactic as a “tactical goal” modeled as Cyber Objective subclass of Objective
- `has_objective` property asserts that its subject intentional act has the objective of achieving a specific purpose
- `Spearphishing_Link_381` has the objective of the object individual `DC_Leaks_Access_73`
- Recommendation: add “`has_objective`” to CCO.



CTT Description, Assumption, Possible Outcome Fields

- Cyber Attack
 - ‘described by’ ‘Descriptive Information Content Entity’
 - ‘has CRA assumption’ ‘Information Content Entity’
 - ‘has possible outcome’ Effect ‘described by’ ‘Descriptive Information Content Entity’

The screenshot displays a web ontology editor interface. On the left, a sidebar shows a hierarchy of object properties, with 'has CRA assumption' selected. The main panel is divided into several sections: 'Annotations' (showing 'has CRA assumption' with its URI), 'Annotations' (listing 'rdfs:label', 'definition', and 'definition_source'), 'Characteristics' (with checkboxes for Functional, Inverse functional, Transitive, and Symmetric), and 'Description' (showing domains and ranges). The 'definition' section contains a detailed description of the property's purpose and the source of the definition.

Annotations **Usage**

Annotations: has CRA assumption

Annotations +

rdfs:label [language: en]
has CRA assumption

definition [language: en]
Relates an Intentional Act to Information Content, which is assumed true by an Agent doing a Assessment of that Act.

definition_source [language: en]
Institute for Defense Analyses, in support of the Cybersecurity/Interoperability Technical Director, Office of the Under Secretary of Defense Research and Engineering (OUSD R&E), Director for Developmental Test, Evaluation, and Assessment (D,DTE&A).

Characteristics **Description: has CRA assumption**

☐ Functional
☐ Inverse functional
☐ Transitive
☐ Symmetric

Domains (intersection) +
● **IntentionalAct**

Ranges (intersection) +
● **InformationContentEntity**

To use the reasoner click Reasoner > Start reasoner ☐ Show Inferences

Attack Timing Classes

Categories of temporal regions of some type of process

The screenshot displays a Semantic Web browser interface. The left pane shows a class hierarchy for 'Maintenance Period', listing various temporal regions. The right pane shows the 'Annotations' tab for the 'Maintenance Period' class, detailing its properties and relationships.

Class hierarchy: Maintenance Period

- one-dimensional temporal region
 - Anytime
 - Data at Rest
 - Data Storage
 - Data Use/Processing
 - Data/Message Transmission
 - Digital Communication
 - Maintenance Period**
 - Mission Data Load
 - Mission Execution/Mission Task
 - Operations Period
 - Platform Configuration-Load
 - Post Mission
 - Post-Mission Download
 - Pre-Mission Planning
 - Software/Firmware/Hardware Update
 - Supply Chain Period
 - Training Event
 - Unspecified
 - Vulnerable Period Before Known Mitigation Applied/Available
 - Vulnerable Period Where Inadequate/No Mitigation is Available

Annotations: Maintenance Period

Annotations:

- rdfs:label** [language: en] Maintenance Period
- CRA time period** [type: xsd:boolean] true
- definition** [language: en] A temporal region of some Act of Maintenance.
- definition_source** [language: en] Institute for Defense Analyses
- is_curated_in_ontology** [type: xsd:anyURI] <http://www.semanticweb.org/OACRA>

Description: Maintenance Period

Equivalent To:

- 'one-dimensional temporal region'
- is_temporal_region_of some ActOfMaintenance

SubClass Of:

- 'one-dimensional temporal region'
- is_temporal_region_of some ActOfMaintenance

General class axioms:

Target and Mitigation Classes in Template

- Cyber Target Classes – hierarchies in Targets Ontology from CTTs
- Mitigations – acts that are objects of ‘has mitigation” property

OPFOR/Engineers (If none are relevant, submit new Target)		Analysis Team	
Cyber Target Superclass	Cyber Target Class	Common Mitigations	Notes to the OACRA Curator
Information_Artifact	Operating System File	Develop and implement patch management plan	
Communication_Entity Cyber_Communication_Entity Information_Artifact Information_Transfer_Entity System_Artifact	Target Superclass Select Cyber Target Superclass. Required.	Unspecified	

OACRA Targets Ontology (1)

- MBCRA/CTT Exercises – source of wide variety of Cyber Targets
- Initial Targets Ontology – used CyberTarget superclass

The screenshot displays the OACRA Targets Ontology interface. On the left, a class hierarchy is shown under the title 'Class hierarchy: oacra:Cyber Target'. The hierarchy starts with 'owl:Thing' and 'obo:Entity', with 'oacra:Cyber Target' highlighted. Below it, several subclasses are listed, including 'oacra:Communication Element', 'oacra:Information Object', and 'oacra:System Element'. On the right, the 'Annotations' panel for 'oacra:Cyber Target' is visible. It includes the following annotations:

- rdfs:label** [language: en]: Cyber Target
- cco:definition** [language: en]: An Entity that is a potential target of a Cyber Attack.
- 'cco:definition source'** [language: en]: Institute for Defense Analyses, in support of the Cybersecurity/Interoperability Technical Director, Office of the Under Secretary of Defense Research and Engineering (OUSD R&E), Director for Developmental Test, Evaluation, and Assessment (D,DTE&A).
- cco:elucidation** [language: en]: Classes included as subclasses of OACRA_Target have instances that have been identified as targets in a previous Cyber Risk Assessment.
- 'oacra:potential Cyber Target'** [type: xsd:boolean]: true

OACRA Targets Ontology (2)

- Revised Targets Ontology

- Eliminated ‘CyberTarget’ class

- Not all instances are actual targets

- Mixes continuants and occurrents since CTTs identify both as targets

- Reorganized former subclass hierarchy

Cyber Target Superclass	Number of Target Classes
System Artifact	95
Information Artifact	92
Communication Entity	46
Information Transfer Entity	14
Cyber Communication	8
Entity	2
All Cyber Target Classes	257

- Distinguished potential targets with class annotation

- ‘CRA target’ a Boolean property – true when an instance of a class has been previously identified during a CRA as a target of a Cyber Attack

Example Target Classes

- Example Target Class with 'CRA target' annotation

The screenshot shows the Protégé OWL editor interface. The top menu bar includes File, Edit, View, Reasoner, Tools, Refactor, Window, and Help. The address bar shows the URL: <http://www.semanticweb.org/OACRA/2023-04-12/Targets>. The breadcrumb trail indicates the current location: entity > continuant > independent continuant > material entity > Communication Entity > Communication Artifact > Communications Device.

The main workspace is divided into two panes. The left pane, titled 'Active ontology x Entities x Individuals by class x DL Query x', shows the class hierarchy for 'Communications Device'. The hierarchy is as follows:

- owl:Thing
 - entity
 - continuant
 - generically dependent continuant
 - independent continuant
 - immaterial entity
 - material entity
 - Agent
 - Communication Entity
 - Communication Artifact
 - Communications Device

The right pane, titled 'Communications Device — http://www.semanticweb.org/OACRA/Targets/Commu...', shows the annotations for the 'Communications Device' class. The annotations are:

 - rdfs:label** [language: en]: Communications Device
 - CRA target** [type: xsd:boolean]: true
 - definition** [language: en]: A Communication Artifact consisting of a hardware device capable of transmitting an analog or digital signal over the telephone, other communication wire, or wirelessly.

The bottom pane, titled 'Description: Communications Device', shows the subclass hierarchy for 'Communications Device'. It lists 'Communication Artifact' as a subclass.

Cyber Vector and Cyber Target Properties

The screenshot displays the Protégé ontology editor interface. The top navigation bar shows the active ontology: **Targets** (<http://www.semanticweb.org/OACRA/2023-04-12/Targets>). Below this, the left sidebar contains a tree view of the ontology structure. Under **Object properties**, the **has cyber attack vector** property is highlighted. The main panel shows the details for this property, including its label, definition, and sources. The **Annotations** tab is active, showing the following information:

- rdfs:label** [language: en]: has cyber attack vector
- definition** [language: en]: A relationship between a CyberAttack and an Entity, which an Agent of the attack intends to be used to gain unauthorized access to a network or system as a result of this attack.
- definition_source** [language: en]: <https://www.fortinet.com/resources/cyberglossary/attack-vector>
- definition_source** [language: en]: <https://www.upguard.com/blog/attack-vector>
- is_curated_in_ontology** [type: xsd:anyURI]: <http://www.semanticweb.org/OACRA/Targets>

The **Description** tab is also visible, showing the domains and ranges of the property:

- Domains (intersection)**: CyberAttack
- Ranges (intersection)**: entity

At the bottom of the interface, a status bar indicates: **No Reasoner set. Select a reasoner from the Reasoner menu** and **Show Inferences** is checked.

Mitigations – acts that are objects of ‘has mitigation’ property

- Mitigations
 - ‘Act of Cyber Threat Mitigation’ range of ‘has mitigation’ property
 - ‘described_by’ ‘Information Content Entity’ ‘has_content’ xsd:String

The screenshot displays a Semantic Web browser interface. On the left, a tree view shows the 'Object property hierarchy: has mitigation'. The hierarchy includes 'owl:topObjectProperty', 'has CRA assumption', 'has possible outcome', 'has_objective', 'inhibited_by' (which contains 'has mitigation'), 'interval_is_after', 'is_cause_of' (which contains 'has result'), 'is_temporal_region_of', and 'RO_0000091'. The 'has mitigation' property is selected and highlighted in blue.

The main panel shows the 'has mitigation' property details. The 'Annotations' tab is active, displaying the following information:

- Annotations: has mitigation**
- Annotations +**
- rdfs:label** [language: en]: has mitigation
- definition** [language: en]: A relationship wherein the effects of Cyber Attack are inhibited by an Act of Cyber Threat Mitigation.
- definition_source** [language: en]: Institute for Defense Analyses, in support of the Cybersecurity/Interoperability Technical Director, Office of the Under Secretary of Defense Research and Engineering (OUSD R&E), Director for Developmental Test, Evaluation, and Assessment (D,DTE&A)

The 'Characteristics' tab is also visible, showing the following properties:

- Functional**
- Inverse function**
- Transitive**
- Symmetric**
- Asymmetric**
- Reflexive**
- Irreflexive**

The 'Description: has mitigation' tab is also visible, showing the following information:

- Equivalent To +**
- SubProperty Of +**: inhibited_by
- Inverse Of +**
- Domains (intersection) +**: CyberAttack
- Ranges (intersection) +**: ActOfCyberThreatMitigation

The bottom status bar indicates: No Reasoner set. Select a reasoner from the Reasoner menu. Show Inferences.

Mitigation Classes – from ATT&CK®

The screenshot displays the Protege ontology editor interface. On the left, the 'Class hierarchy' pane shows a tree structure under 'Account Use Policies', with 'Act of Cyber Threat Mitigation' expanded to show various mitigation classes. 'Account Use Policies' is highlighted. The main pane shows the 'Annotations' tab for 'Account Use Policies'. It lists several annotations: 'rdfs:label' (Account Use Policies), 'definition' (Configure features related to account use like login attempt lockouts, specific login times, etc. An Act of Cyber Threat Mitigation.), 'definition source' (https://attack.mitre.org/mitigations/M0936 and https://attack.mitre.org/mitigations/M1036), and 'is curated in ontology' (http://www.ontologyrepository.com/CommonCoreOntologies/Domain/MAMO). Below the annotations, the 'Description' tab shows 'Account Use Policies' as a subclass of 'Act of Cyber Threat Mitigation'. It also lists two 'inhibits some' relationships with specific enterprise techniques. The bottom status bar indicates 'To use the reasoner click Reasoner > Start reasoner' and 'Show Inferences'.

Annotation properties | Datatypes | Individuals

Classes | Object properties | Data properties

Class hierarchy: Account Use Policies

Asserted

- Act of Inhibiting
 - Act of Cyber Threat Mitigation
 - Access Management
 - Account Use Policies
 - Active Directory Configuration
 - Antivirus/Antimalware
 - Application Developer Guidance
 - Application Isolation and Sandboxing
 - Attestation
 - Audit
 - Authorization Enforcement
 - Behavior Prevention on Endpoint
 - Boot Integrity
 - Code Signing
 - Communication Authenticity
 - Credential Access Protection
 - Data Backup
 - Data Loss Prevention
 - Deploy Compromised Device Detectic
 - Disable or Remove Feature or Prograr
 - Do Not Mitigate
 - Encrypt Network Traffic
 - Encrypt Sensitive Information
 - Enterprise Policy
 - Environment Variable Permissions

Annotations | Usage

Annotations: Account Use Policies

Annotations +

- rdfs:label** [language: en] [in MAMO]
Account Use Policies
- definition** [language: en] [in MAMO]
Configure features related to account use like login attempt lockouts, specific login times, etc. An Act of Cyber Threat Mitigation.
- definition source** [language: en] [in MAMO]
<https://attack.mitre.org/mitigations/M0936>
- definition source** [language: en] [in MAMO]
<https://attack.mitre.org/mitigations/M1036>
- is curated in ontology** [type: xsd:anyURI] [in MAMO]
<http://www.ontologyrepository.com/CommonCoreOntologies/Domain/MAMO>

Description: Account Use Policies

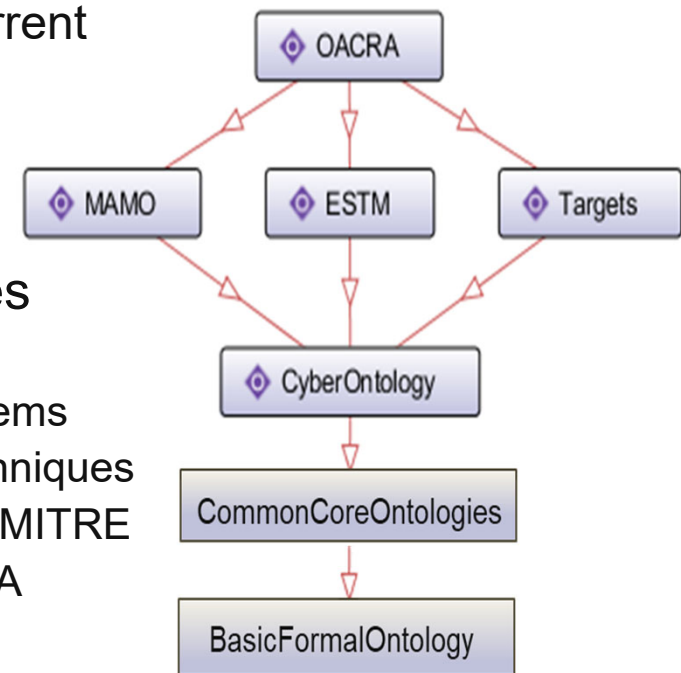
SubClass Of +

- 'Act of Cyber Threat Mitigation'
- inhibits some
(('Brute Force Enterprise Technique' or 'Credential Stuffing Enterprise Technique' or 'Multi-Factor Authentication Request Generation Enterprise Technique' or 'Password Guessing Enterprise Technique' or 'Password Spraying Enterprise Technique'))
- inhibits some
(('External Remote Services ICS Technique' or 'Valid Accounts ICS Technique'))

To use the reasoner click Reasoner > Start reasoner ☐ Show Inferences

OACRA Component Ontologies

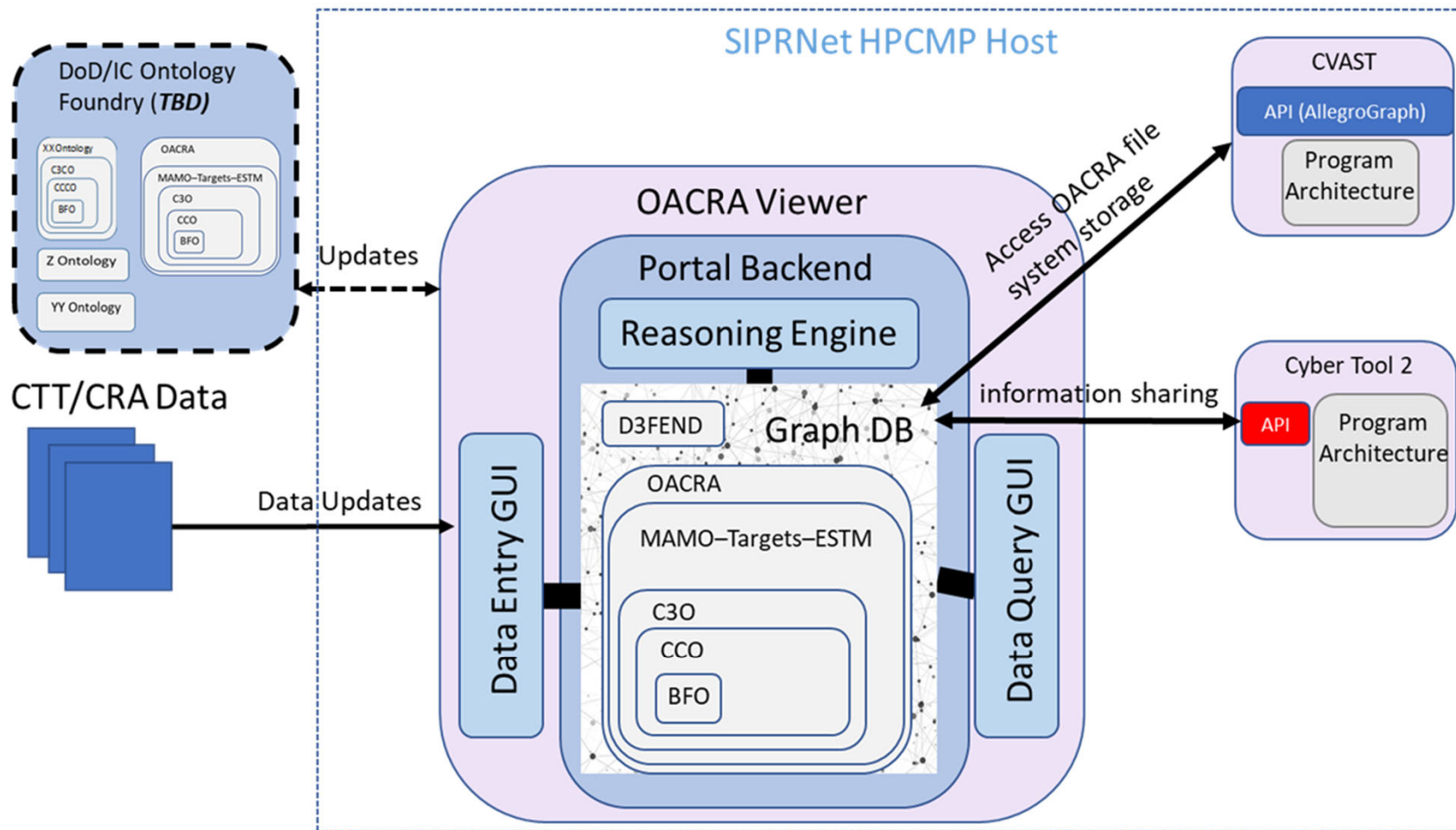
- MITRE ATT&CK Matrix Ontology (MAMO)
 - Includes ATT&CK Tactics, Techniques, Mitigations, and Groups
 - Originally developed by J. Reynolds, lead of C3O WG
 - Updated several times by OACRA, but not current
- Targets Ontology
 - Based on target categories used in CTT exercises
 - Developed by IDA with TRMC and OACRA WG
- Embedded Systems Tactics and Techniques Matrix Ontology (ESTM)
 - Focused on subsystems/components of larger systems
 - Consistent with ATT&CK approach to Tactics & Techniques
 - ESTM matrix document developed by M. Zuniga at MITRE
 - ESTM ontology developed from ESTM matrix by IDA
- Common Core Cyber Ontology (C3O)
 - Imported whole with all of CCO and BFO
 - Minor modification – to add 'has objective' property



OACRA Viewer and Graph Database

- OACRA Graph Database
 - Captures Cyber Attack data from MBCRAs of DoW systems and their components
 - Uses triple store based on OACRA, the ontology
 - Sponsor chose DoW HPC Modernization Program (HPCMP) for hosting the OACRA Graph DB and OACRA Viewer
 - HPCMP provides no-cost hosting unclassified and classified
 - IDA coordinated with contractors who implemented it
- OACRA Viewer
 - User Interface to OACRA Graph Database
 - Auto-loads data from OACRA template spreadsheets
 - Supports user access to all data through multiple filters

OACRA Viewer



User will leverage standard framework/ontology and query database for similar cyber targets OR types of attacks

OACRA Integration with D3FEND

- Anticipated Funding Limitations Required Other Support
 - Cybersecurity and Interoperability Deputy Director, DTE&A
 - 2024 - DTE&A unlikely to be able to fund OACRA updates
 - Need other sources to help update MAMO and Targets ontologies
 - MITRE D3FEND ontology maintains an up-to-date ATT&CK Ontology
 - D3FEND ontology has extensive mitigation techniques and digital artifacts
- Initiated Transition to Integrate OACRA with D3FEND
 - D3FEND-CCO development initiated by MITRE for compatibility
 - D3FEND mitigations integrated into OACRA Graph DB
 - IDA prepared mappings between D3FEND and OACRA/C3O
- Changes to DTE&A Roles required further cutbacks
 - Ongoing support for OACRA sought and obtained from external Agency
 - OACRA plans reconfigured to integrate with D3FEND-CCO

OACRA Way-Ahead

- Promote use of current OACRA Viewer while exploring alternatives
- Shift from use of OACRA with C3O to use D3FEND-CCO
 - Used by industry to model sanitized cyber-attacks for reuse in MBCRAs
 - DoW to continue to model sanitized cyber-attacks for reuse at the appropriate classification levels
 - Support integration with model-based systems engineering for improving earlier, automated cyber analysis of DoW systems in digital engineering ecosystems
 - Sustained updates and integration using MITRE ATT&CK in D3FEND-CCO
- Pursue Use of D3FEND's CAD tool in DoW classified environments to model cyber-attacks for MBCRAs
- Creates Challenge for Integration with Systems using C3O
 - Mitigation by thorough mappings between C3O and D3FEND-CCO

For More Information on OACRA

- **Final Task Report: Ontology for Attacks in Cyber Risk Assessment (OACRA)**
 - More detailed report on OACRA project
 - Publication Date: September 2025
 - Pagination: 79
 - Download from Defense Technical Information Center (DTIC)
 - <https://www.dtic.mil/document;accessionNumber=AD1342540;type=DTICTR;searchText=OACRA>
 - Contact authors
- **Author Contact Information**
 - Brian A. Haugh – bhaugh@ida.org
 - Steven P. Wartik – swartik@ida.org

BACKUP SLIDES

OACRA WG DoW Component Engagements

- Automated Cyber Survivability (ACS) Cyber Survivability Attributes (CSA) Tool
 - Air Force Research Laboratory (AFRL) program
 - Enables automated cyber survivability of systems in support of AF missions
 - Enables cyber failover without loss of critical information (2 minutes or faster)
- CALDERA
 - A MITRE tool
 - Simulates the goals and actions of a real adversary
 - Uses a system of mappers, mappings, and exporters to generate CRA data using a target ontology (e.g., OACRA) from CALDERA reports
- Cyber Vulnerability Assessment Tool (CVAST)
 - Digital Engineering Decision Support System for cyber risk assessment and engineering analysis of Navy systems
 - Identifies quantifiable cyber risk to mission
 - Can consume data from OACRA Viewer
- Mission-based Risk Assessment Process for Cyber (MRAP-C)
 - Air Force program for MBCRA
 - Integrates MBCRAs and engineering cyber assessment activities
 - Iterative MBCRAs build upon previous assessments

OACRA 'has text value' property

- Simplifies attaching text to describe Information Content

The screenshot displays the Protégé ontology editor interface. On the left, a sidebar shows a hierarchy of properties under 'owl:topDataProperty'. The 'has text value' property is highlighted in blue. The main area shows the details for this property, including its URI, annotations, and domain/range information.

has text value — http://www.semanticweb.org/OACRA/has_text_value

Annotations: has text value

rdfs:label [language: en]
has text value

definition [language: en]
A relationship between either an Information Bearing Entity or an Information Content Entity and a string that represents some of the information contained in or expressed by the entity.

definition source [language: en]
Institute for Defense Analyses, in support of the Cybersecurity/Interoperability Technical Director, Office of the Under Secretary of Defense Research and Engineering (OUSD R&E), Director for Developmental Test, Evaluation, and Assessment (D,DTE&A).

is curated in ontology [language: en]
<http://www.semanticweb.org/OACRA>

Description: has text value

☐ Functional

Domains (intersection) +
• 'Information Bearing Entity' or 'Information Content Entity'

Ranges +
• xsd:string

To use the reasoner click Reasoner > Start reasoner ☐ Show Inferences